Course intro & what is R?

Day 1 - Monday, May 18, 2026

Introductions

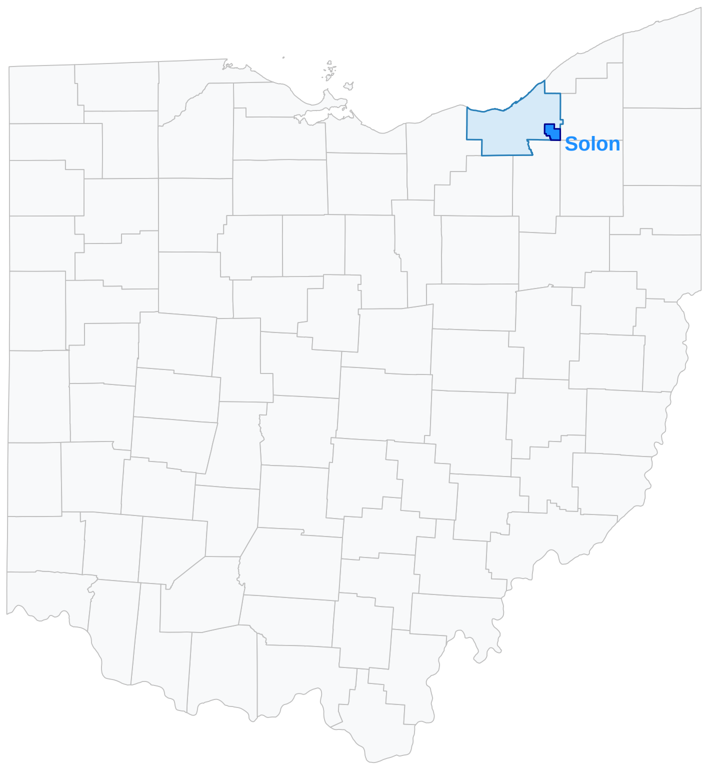
Instructor: Charles Costanzo

- Statistics Ph.D. student
- Hometown: Solon, OH
- Some of my favorite things: cats, coffee, R (unironically)

Solon, Ohio

```r
# Install these first if you don't have them:
# install.packages(c("sf", "tigris"))
library(ggplot2)
library(sf)
library(tigris)
library(dplyr)

options(tigris_use_cache = TRUE)

# Fetch spatial data
ohio_counties <- counties(state = "OH", cb = TRUE, progress_bar = FALSE)
ohio_places <- places(state = "OH", cb = TRUE, progress_bar = FALSE)

# Isolate areas
cuyahoga_county <- ohio_counties %>% filter(NAME == "Cuyahoga")
solon_city <- ohio_places %>% filter(NAME == "Solon")

# Build the map
ggplot() +
```

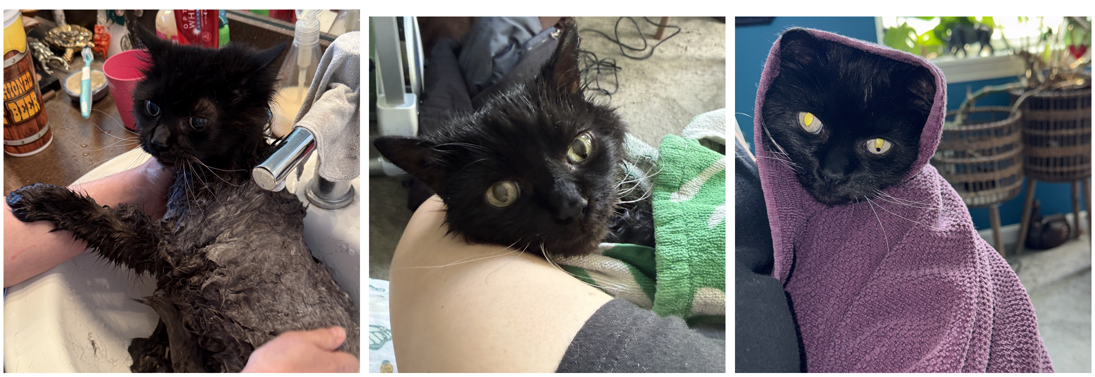
```
geom_sf(data = ohio_counties,
        fill = "#f8f9fa", color = "grey", linewidth = 0.3) +
geom_sf(data = cuyahoga_county,
        fill = "#D6EAF8", color = "#2980B9", linewidth = 0.5) +
geom_sf(data = solon_city,
        fill = "dodgerblue", color = "darkblue", linewidth = 0.5) +
geom_sf_text(data = solon_city, aes(label = NAME),
             size = 5, fontface = "bold", color = "dodgerblue",
             nudge_y = -0.05, nudge_x = 0.25) +
theme_void()
```



Meet My Cats!

Babuh (left) and Tartarus (right).

More Realistic Cat Photos...

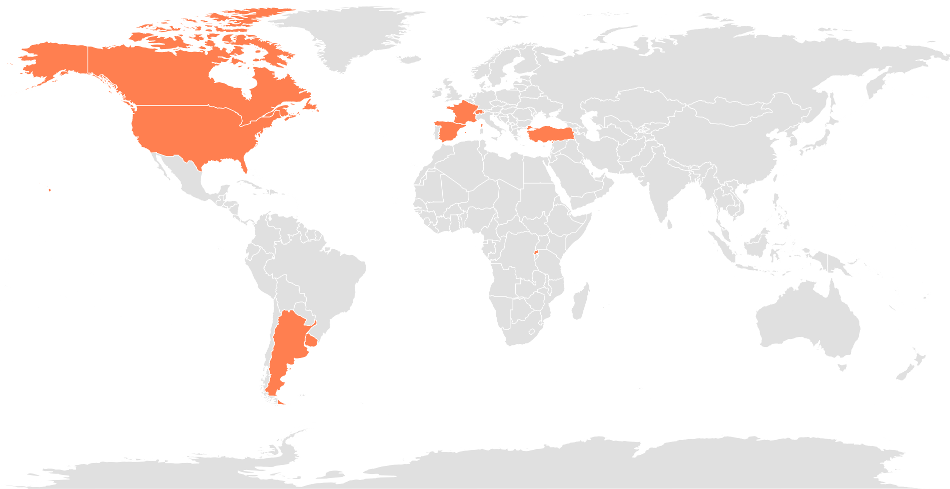


Babuh: soaked and indignant, mid-bath, then wrapped in a towel.

Countries I've Been To

```r
library(ggplot2)
library(maps)
library(dplyr)
```

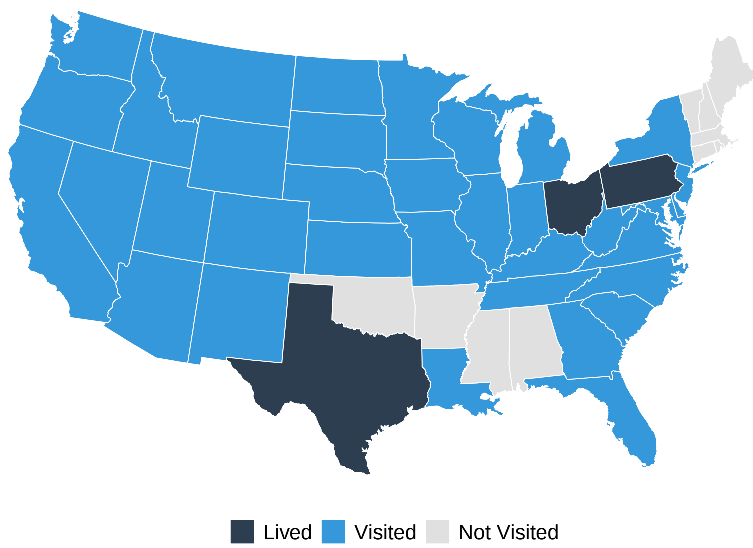
```r
world_map <- map_data("world")

visited_countries <- c("Canada", "USA", "Argentina", "Uruguay",
                       "Switzerland", "Rwanda", "Spain", "France", "Turkey")

world_map <- world_map %>%
  mutate(visited = ifelse(region %in% visited_countries, "Visited", "Not Visited"))

ggplot(world_map, aes(x = long, y = lat, group = group, fill = visited)) +
  geom_polygon(color = "white", linewidth = 0.2) +
  scale_fill_manual(values = c("Not Visited" = "#e0e0e0", "Visited" = "#FF7F50")) +
  theme_void() +
  theme(legend.position = "none")
```



States I've Lived In & Visited

```r
# Note: install.packages("mapproj") if needed
library(ggplot2)
library(maps)
library(dplyr)
library(mapproj)

us_map <- map_data("state")
```

```r
lived_in <- tolower(c("Ohio", "Pennsylvania", "Texas"))

# Note: "District of Columbia" matches the map data; "Washington DC" wouldn't
visited <- tolower(c("Washington", "Oregon", "California", "Nevada", "Idaho",
                     "Montana", "Arizona", "New Mexico", "Colorado", "Wyoming",
                     "North Dakota", "South Dakota", "Kansas", "Iowa",
                     "Nebraska", "Minnesota", "Wisconsin", "Michigan", "Indiana",
                     "Illinois", "Kentucky", "Tennessee", "Louisiana", "Florida",
                     "Georgia", "North Carolina", "South Carolina", "Virginia",
                     "District of Columbia", "Maryland", "Delaware", "New Jersey",
                     "New York", "Missouri", "Utah", "West Virginia"))

us_map <- us_map %>%
  mutate(status = case_when(
    region %in% lived_in ~ "Lived",
    region %in% visited ~ "Visited",
    TRUE ~ "Not Visited"
  )) %>%
  mutate(status = factor(status, levels = c("Lived", "Visited", "Not Visited")))

ggplot(us_map, aes(x = long, y = lat, group = group, fill = status)) +
  geom_polygon(color = "white", linewidth = 0.3) +
  scale_fill_manual(values = c("Lived" = "#2C3E50",
                               "Visited" = "#3498DB",
                               "Not Visited" = "#E0E0E0")) +
  coord_map("albers", lat0 = 39, lat1 = 45) +
  theme_void() +
  theme(legend.position = "bottom",
        legend.title = element_blank(),
        legend.text = element_text(size = 14))
```



Icebreaker : Share Your...

- Name
- Major
- Programming Experience
- A Fun Fact (Favorite Song, Color, Food, etc.)

Today: Monday May 18

Before class

- Read the syllabus (Click here).
- (Optional): Read *Data Computing* §1.1–1.5 (Click here).

Plan for today

- Course logistics
- Introduction to R and RStudio
- Tidy data
- R Demo

Helpful links

- Syllabus · AI policy · Schedule · *Data Computing* §1

Course logistics

Course Info

Summer Session 1: May 18, 2026 – June 26, 2026

- **Class time:** M/T/W/Th/F, 9:35 a.m. to 10:50 a.m.
- **Class location:** Huck Life Sciences Building 007
- **Instructor:** Charles Costanzo
  - Email: cgc5478@psu.edu
  - Office Hours: M/W/Th/F: 10:50-11:50 a.m. in Huck 005 (or on Zoom by appt)
- **Teaching Assistant (TA):** Xinyue Wang
  - Email: xpw5228@psu.edu

Learning Goals

This course will teach you how to:

- **Use R**: Program in the R *language* effectively and use the R *software environment* to efficiently compute with data.
- **Wrangle Real Data**: Access, clean, join, and format data from various sources.
- **Visualize Information**: Create graphical and descriptive summaries of data.
- **Build Reproducible Workflows**: Write well-documented code using tools like RStudio, RMarkdown, and Git/GitHub.

No previous programming background is required for this course.

! This is **not** a course on general purpose programming.

Course logistics

- **Six weeks is short.** Daily pace is faster than a 15-week course. Falling a week—or even a day—behind is more serious than spring/fall courses.
- **Attendance matters.** Lots of in-class graded work (activities, quizzes, oral checkpoints).
- **Timely communication is critical.** Reach out to me **as soon as possible** if something comes up that necessitates you missing class, having a deadline extended, etc.

Table 1: **Assignment weights**

Readings	Quizzes	In-class activities	Homework	Exam	Course project
10%	15%	25%	10%	20%	20%

Table 2: **Grading scheme**

A	A-	B+	B	B-	C+	C	D	F
93%	90%	87%	83%	80%	77%	70%	60%	below 60%

AI policy

- **Open assessments (Homework & Project):** AI tools are permitted **with mandatory disclosure**.
- **Closed-book, in-person work (Quizzes, Exam, Activities, Oral Checkpoints):** AI tools are **strictly prohibited**.
- **Disclosure is graded (1-2 points).**
- You must be able to explain any code you submit, which I will verify during in-class oral project checkpoints.
- Roughly 60% of your grade comes from verified in-person work where AI cannot be used.

Course Readings (Perusall)

- All assigned textbook readings are hosted on **Perusall**, a collaborative e-reading platform.
- **How grading works:** You earn credit through holistic engagement (reading the entire text, spending active time, leaving high-quality comments, and replying to classmates).
- **No access code needed!** You must access readings *directly through Canvas links* to automatically authenticate your Penn State account.

❗ Let's try it together right now:

1. Log into our Canvas course.
2. Click the "Perusall" tab on the left.
3. Click the link for **DC Section 1 (§1.1–1.5)**.

What is R?

What is R?

R is a **programming language** *and* a **software environment** for statistical computing and graphics.

- Designed by statisticians for the purpose of data analysis.
- First version created in August 1993 (almost 33 years ago).
- Has grown rapidly in popularity and is widely used in academia and industry by statisticians, data analysts, and data scientists alike.
- Enormous number of user-contributed **packages**, which are add-ons that extend the functionality of R.
  - Currently the Comprehensive R Archive Network (CRAN) package repository features 23,664 available packages.[1]

Some Examples of Said R Packages

[1] As of May 17, 2026.

Hex stickers from the rstudio/hex-stickers repository.

R vs. RStudio

- **R** is a computer language that performs computations based on human-written instructions.
- **RStudio** is an integrated development environment (**IDE**) for R programming. It combines the R console with supplementary tools for more efficient R programming.

*Left: **R** (don't open this). Right: **RStudio** (open this).*

Tidy data

Tidy Data ≠ Neat Data

> This table is easy for humans to read. What might make it tricky for a computer to read?

	Week Of 05-04-2026 to 05-11-2026				Yearly Total 2026			
	To Democratic		To Republican		To Democratic		To Republican	
COUNTY	R	OTH	D	OTH	R	OTH	D	OTH
ADAMS	5	4	4	3	71	93	74	102
ALLEGHENY	47	125	23	31	786	1,323	810	389
ARMSTRONG	2	0	1	1	23	28	65	26
BEAVER	2	15	3	1	77	127	150	55
BEDFORD	0	1	0	1	13	14	21	16
BERKS	15	37	14	16	337	494	422	335
BLAIR	5	7	0	3	60	83	62	52
BRADFORD	1	2	1	2	32	27	40	45
BUCKS	32	94	15	20	626	1,054	550	557
BUTLER	3	5	0	3	131	170	101	111
CAMBRIA	1	2	3	0	82	71	174	62
CAMERON	0	0	0	0	1	1	7	0
CARBON	5	11	4	5	34	70	54	43
CENTRE	2	14	3	4	88	130	48	75
PHILADELPHIA	60	313	66	29	1,076	3,163	1,173	489
VENANGO	2	2	0	0	28	35	30	27
WARREN	0	0	1	1	24	32	15	24
WASHINGTON	5	7	7	3	112	142	188	111
WAYNE	1	1	3	2	22	35	41	39
WESTMORELAND	8	20	12	8	211	238	302	168
WYOMING	0	2	3	0	22	22	27	17
YORK	24	60	10	22	415	656	325	393
Totals:	**512**	**1,348**	**334**	**322**	**9,823**	**15,604**	**9,631**	**7,295**

Figure 1: A spreadsheet showing party change applications in PA. Source: PA Dept. of State (May 4–11, 2026).

Two simple rules for Tidy Data:

1. **Consistent Cases (Rows)**: Each row must represent a single, distinct instance of your unit of observation.
2. **Uniform Variables (Columns)**: Every column contains the same type of value.

PA Voter Table - Tidied Up

```
# 1. Load data and provide clean column names to handle the nested headers
raw_data <- readxl::read_excel("data/currentvotestats.xlsx",
                       sheet = "Party-to-Party(2026)",
                       skip = 3, # Skip the human-readable headers
                       col_names = c("county",
                                     "week_dem_R", "week_dem_OTH",
                                     "week_rep_D", "week_rep_OTH",
                                     "year_dem_R", "year_dem_OTH",
                                     "year_rep_D", "year_rep_OTH"))

# 2. Clean and Pivot
tidy_voter_data <- raw_data %>%
  filter(!is.na(county), county != "Totals:") %>%
  pivot_longer(
    cols = -county,
    names_to = c("period", "target_party", "origin_party"),
    names_sep = "_",
    values_to = "count"
  )

knitr::kable(
  head(tidy_voter_data %>% filter(county == "CENTRE")),
  caption = "First six rows of the tidied PA voter party-change data, filtered to Centre Cou
)
```

Table 3: First six rows of the tidied PA voter party-change data, filtered to Centre County. Each row is one combination of period, target party, and origin party with its corresponding count.

county	period	target_party	origin_party	count
CENTRE	week	dem	R	2
CENTRE	week	dem	OTH	14
CENTRE	week	rep	D	3
CENTRE	week	rep	OTH	4

county	period	target_party	origin_party	count
CENTRE	year	dem	R	88
CENTRE	year	dem	OTH	130

Demos

Demo: what R can do

```
library(tidyverse)
library(palmerpenguins)
```

```
Attaching package: 'palmerpenguins'


The following objects are masked from 'package:datasets':

    penguins, penguins_raw
```

```
# What does the data look like?
glimpse(penguins)
```

```
Rows: 344
Columns: 8
$ species           <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island            <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm    <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
$ bill_depth_mm     <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
$ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
$ body_mass_g       <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
$ sex               <fct> male, female, female, NA, female, male, female, male~
$ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

```
# Quick summary by species
penguins |>
  group_by(species) |>
  summarize(
    n             = n(),
```

```r
    mean_mass_g  = mean(body_mass_g,    na.rm = TRUE),
    mean_bill_mm = mean(bill_length_mm, na.rm = TRUE)
  )
```

```
# A tibble: 3 x 4
  species       n mean_mass_g mean_bill_mm
  <fct>     <int>       <dbl>        <dbl>
1 Adelie      152       3701.         38.8
2 Chinstrap    68       3733.         48.8
3 Gentoo      124       5076.         47.5
```

```r
# A nice plot
penguins |>
  ggplot(aes(x = bill_length_mm, y = body_mass_g, color = species)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Bill length vs body mass, by species",
    x = "Bill length (mm)",
    y = "Body mass (g)"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

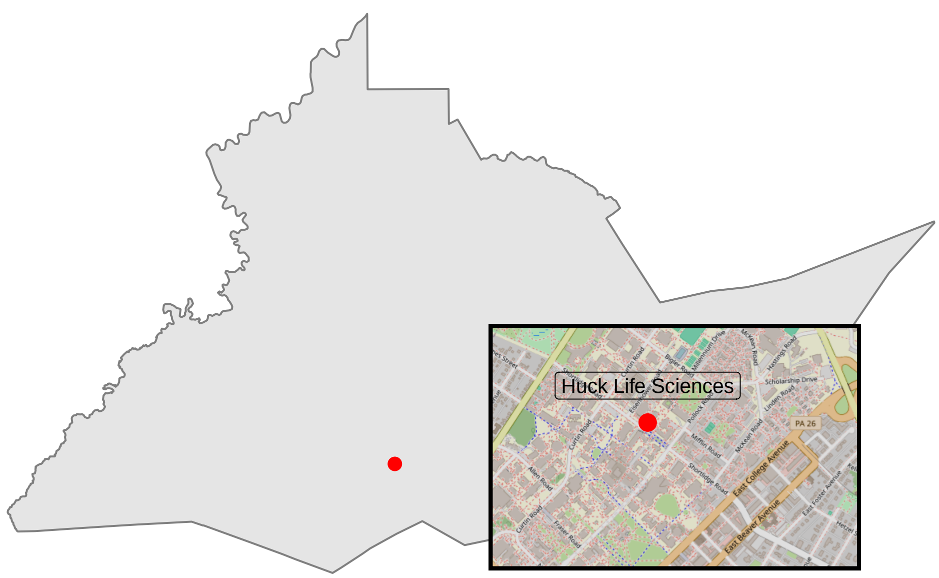
```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```

Demo: a campus map

```r
library(tidyverse)
library(sf)
library(tigris)
library(ggspatial)
library(patchwork)

options(tigris_use_cache = TRUE)
centre_county <- counties(state = "PA", cb = TRUE, progress_bar = FALSE) |>
  filter(NAME == "Centre")

huck_building <- tibble(
  lon = -77.86146,
  lat = 40.80100,
  name = "Huck Life Sciences"
) |>
  st_as_sf(coords = c("lon", "lat"), crs = 4326)

map_main <- ggplot() +
  geom_sf(data = centre_county, fill = "gray90", color = "gray50",
```

```
          linewidth = 0.5) +
  geom_sf(data = huck_building, color = "red", size = 3) +
  theme_void() +
  labs(title = "Centre County, Pennsylvania")

map_inset <- ggplot() +
  annotation_map_tile(type = "osm", zoom = 15, progress = "none") +
  geom_sf(data = huck_building, color = "red", size = 4) +
  geom_sf_label(data = huck_building, aes(label = name), nudge_y = 0.0015) +
  coord_sf(
    xlim = c(-77.870, -77.850),
    ylim = c(40.795, 40.805),
    expand = FALSE,
    crs = 4326
  ) +
  theme_void() +
  theme(panel.border = element_rect(color = "black", fill = NA,
                                    linewidth = 2))

map_main +
  inset_element(map_inset, left = 0.45, bottom = 0.05,
                right = 0.95, top = 0.45)
```

Centre County, Pennsylvania



Setup checklist — due by Wednesday

Install R and RStudio

- R: https://cloud.r-project.org
- RStudio: https://posit.co/download/rstudio-desktop

> ⚠ Mac users
>
> Install **XQuartz** too — some packages we'll use later in the course need it.

Configure RStudio

- `Tools > Global Options... > General`
- Restore .RData into workspace at startup: **Unchecked**
- Save workspace to .RData on exit: **Never**

Assignments due Wednesday 5/20

- HW 0: Setup confirmation

Wrap-up & looking ahead

Tomorrow: A tour of RStudio, R as a calculator, and creating your first objects

Upcoming Deadlines

- **Tue 5/19, before class** — DC §2.1–2.8
- **Wed 5/20, 11:59 p.m.** — HW 0: Setup
- **Fri 5/22, in class** — Quiz 1
  - Quiz 1 Practice
- **Fri 5/22, 11:59 p.m.** — HW 1: Basic R & Plots

Reach out

Stuck? Office hours, or email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu).

Reference

Syllabus · AI policy · Schedule · *Data Computing*